

CSCE 580: Intro to AI — Quiz 1 Responses

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Submission Logistics

All code, data, and this report are under `Quiz1/` in the GitHub repository. I am submitting two notebooks: `Q2-code.ipynb` and `Q3-code.ipynb`. Figures exported by the notebooks are placed next to this `.tex` file and included below.

1 Q1: About data for AI (20 pts)

(a) What is open data? Include one example I produce

- **Definition.** *Open data* is information that anyone can freely access, use, modify, and share for any purpose, subject only to minimal restrictions that preserve openness (typically attribution and/or share-alike). To be truly useful, it should be machine-readable, documented, and licensed for reuse.
- **Example I produce.** In this repo, I publish non-identifying analysis artifacts (e.g., figure PNGs and aggregated keyword tables) under an open license so classmates can reuse them. I can also release an anonymized list of top resume keywords as open data for aggregate analysis.

(b1) Two reasons attributes may be missing

- **Parsing / ingestion issues.** PDFs and DOCX files use heterogeneous layouts; text embedded as images or unusual encodings often leads to dropped or garbled fields (e.g., dates).
- **Source incompleteness.** Authors omit details (e.g., months for employment dates) or data entry forms leave fields optional; merging heterogeneous sources can also misalign field names.

(b2) Two ways to proceed (with assumptions and risks)

- **Impute.** For sporadic gaps, fill numeric values with the median and categorical values with the mode. *Assumption:* missingness is roughly random (MCAR/MAR). *Risk:* may bias distributions, shrink variance, and give overconfident models if values are MNAR.
- **Analyze robustly with omission thresholds.** Drop rows/columns above a missingness threshold (e.g., $>30\%$) and log the removals. *Assumption:* removed items are not systematically different. *Risk:* selection bias and reduced sample size.

2 Q2: Resume Analysis (25 pts)

All code and figures are generated by `Q2-code.ipynb`. The workflow:

1. Ingest my resume and the class resume folder (TXT/PDF/DOCX) and tokenize to words (lowercased).
2. Remove English stopwords (Sebleier list with an offline fallback in the notebook).
3. Plot top-20 words before/after stopwords removal for my resume and for the class aggregate.
4. Compare my *specific words* (after stopwords) to the class' *specific words* to find tokens unique to me.

Figures & short observations

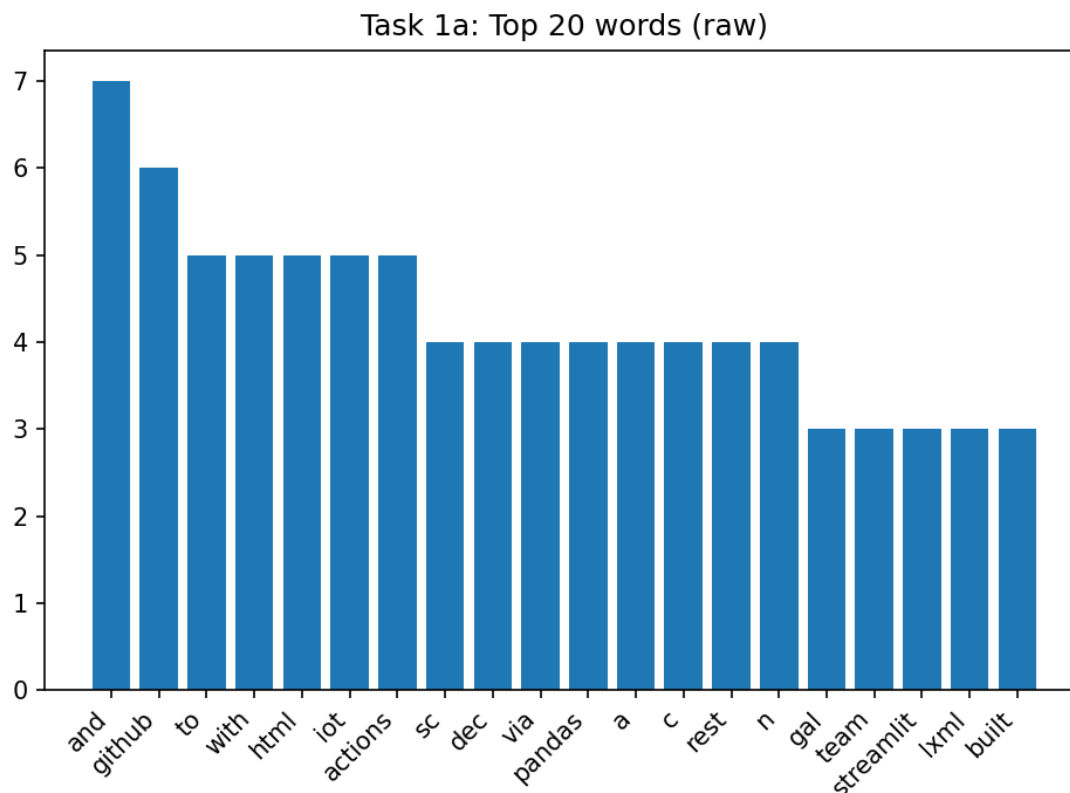


Figure 1: My resume — Top 20 words (raw).

Interpretation (concise). After stopwords removal, my resume surfaces technical tokens such as *github*, *iot*, *pandas*, *rest*, *streamlit*, and *docker*, indicating hands-on software/IoT work. The class aggregate emphasizes *data*, *computer*, *science*, *software*, *team*, and *programming*, consistent with CS/engineering majors. Tokens present for me but rare in the class (e.g., *github*, *streamlit*, *docker*, *iot*, *esp*) highlight differentiation.

3 Q3: Fire Station Data — EDA and Unsupervised Learning (55 pts)

All code and exact values appear in `Q3-code.ipynb`. Below I summarize the approach and findings; figures are exported from the notebook.

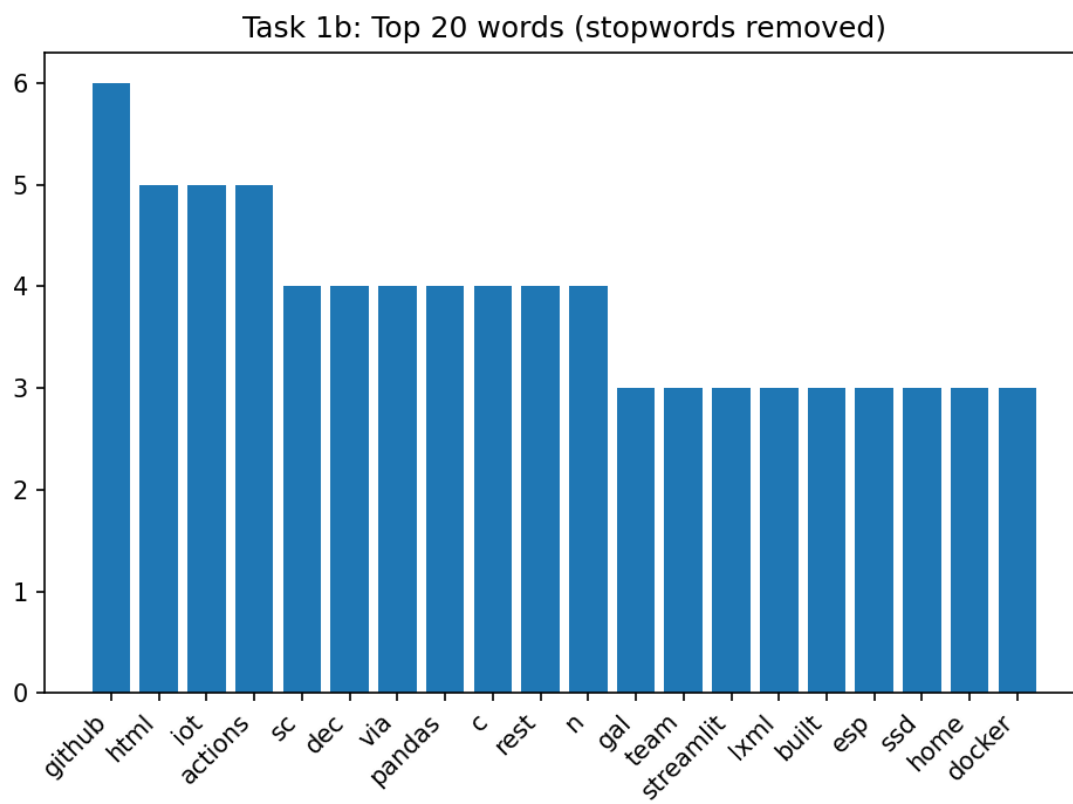


Figure 2: My resume — Top 20 words (stopwords removed).

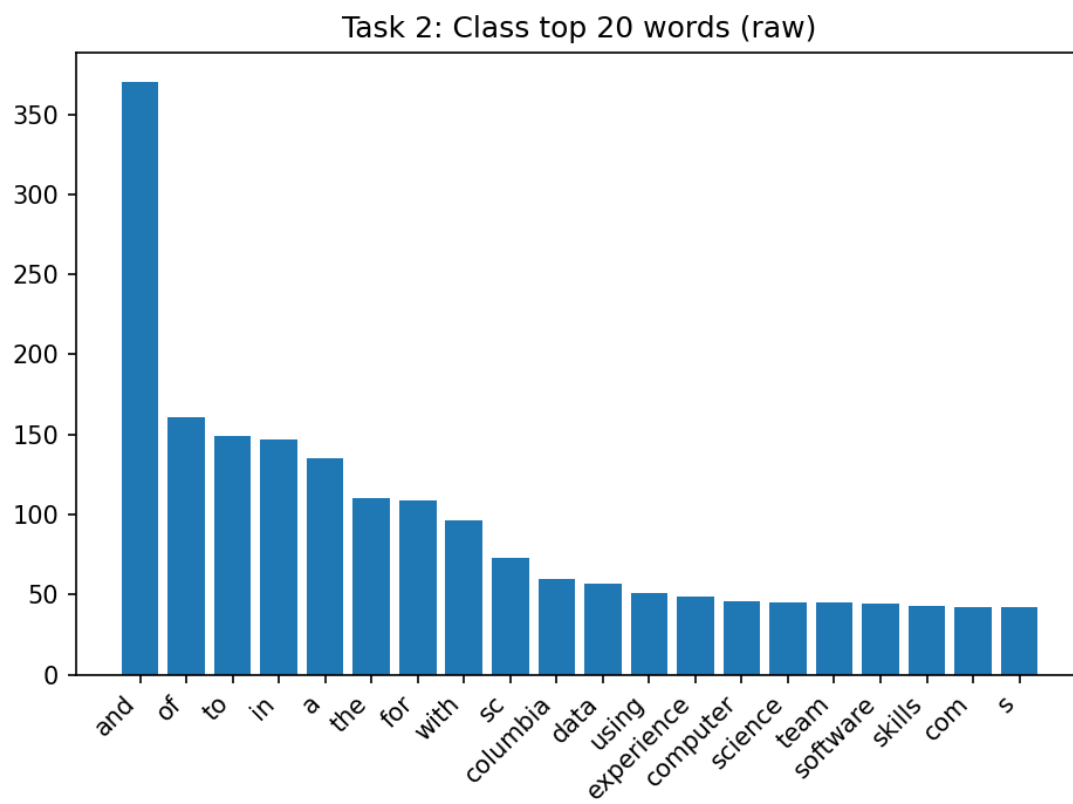


Figure 3: Class resumes — Top 20 words (raw).

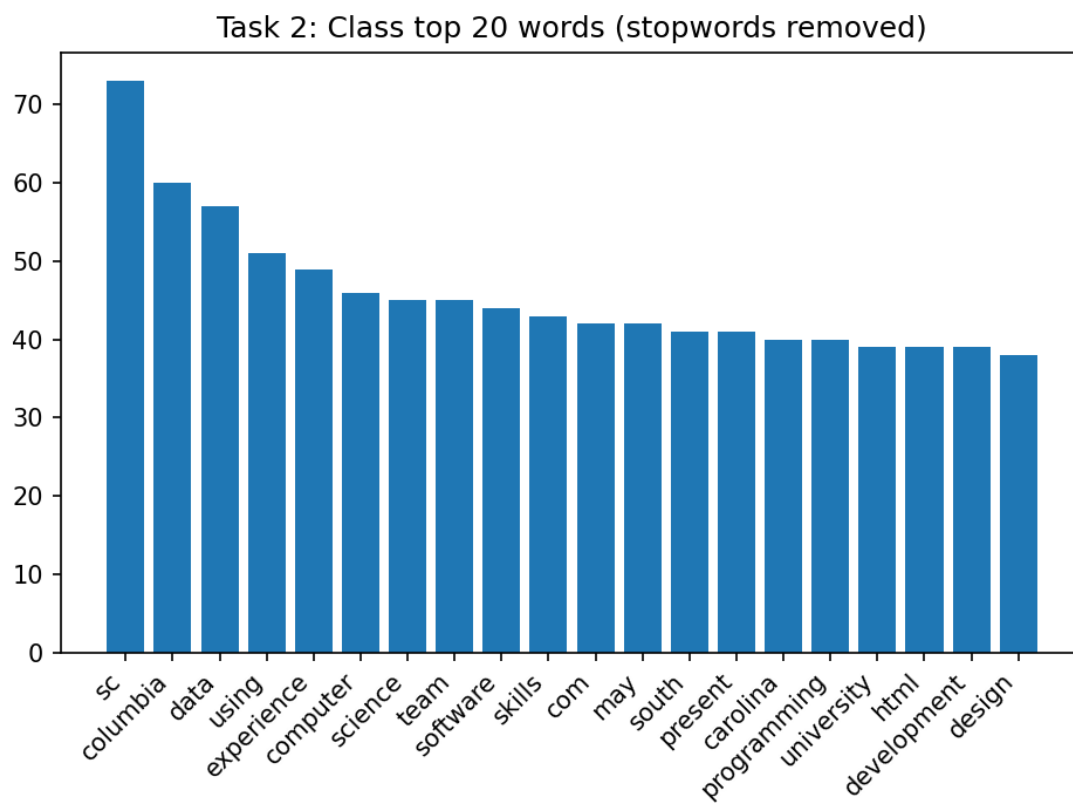


Figure 4: Class resumes — Top 20 words (stopwords removed).

(a) Data issues and cleaning (15 pts)

Goals: determine date ranges, quantify missingness, standardize formats, and produce a clean dataframe for analysis.

- **Date range.** Parsed `CREATED` and `CLOSED` timestamps and reported min/max (see notebook cell *CREATED/CLOSED range*).
- **Missingness.** Computed % missing by column; primary issues were occasional null timestamps/shift values.
- **Detected issues.** Mixed datetime formats; free-text lists of responding units; absent/ambiguous shift labels in some rows.
- **Resolutions.** Robust datetime parsing with multiple formats and coercion; `UNITS_COUNT` from splitting token lists; `SHIFT` derived from the hour if missing (A: 07–14:59, B: 15–22:59, C: 23–06:59); median imputation for sparse numeric gaps; text filled with “Unknown”; preserved a unique ID.

(b) Exploratory data analysis (20 pts)

- **Average time to resolve a call.** Computed as `CLOSED - CREATED` in minutes (`DURATION_MIN`); see notebook output for the exact value.
- **Average number of units per alarm.** Mean of `UNITS_COUNT` across incidents.
- **Busiest shift.** Count incidents per `SHIFT` (A/B/C).
- **Workload matrix.** Pivot of incident counts by day-of-week (columns) and hour (rows), with row/column totals.

(c) Unsupervised learning (20 pts)

Features: `DURATION_MIN`, `UNITS_COUNT`, `HOURL`, and encoded `DOW`. All features are standardized before clustering.

Methods compared. I evaluate **KMeans** across $k \in \{2, \dots, 6\}$ and an **Agglomerative** model with the same k as the best KMeans. The selection criterion is the highest silhouette score; ties break by interpretability.

Result and interpretation. The higher-silhouette method is selected in the notebook and assigned to `CLUSTER`. Cluster means (below) support plain-language labels such as: *short/low-unit routine calls* vs. *longer/multi-unit incidents*, with time-of-day effects captured by `HOURL` and `DOW`.

Reproducibility & Integrity

- Figures were generated by the notebooks using `matplotlib` (one chart per figure, no custom colors).
- External resource: Sebleier English stopwords list; the notebook includes an offline fallback.
- Paths are configurable at the top of each notebook; running top-to-bottom reproduces all results.
- I used AI tools only for drafting and editing; the analyses, code, and interpretations are my own.

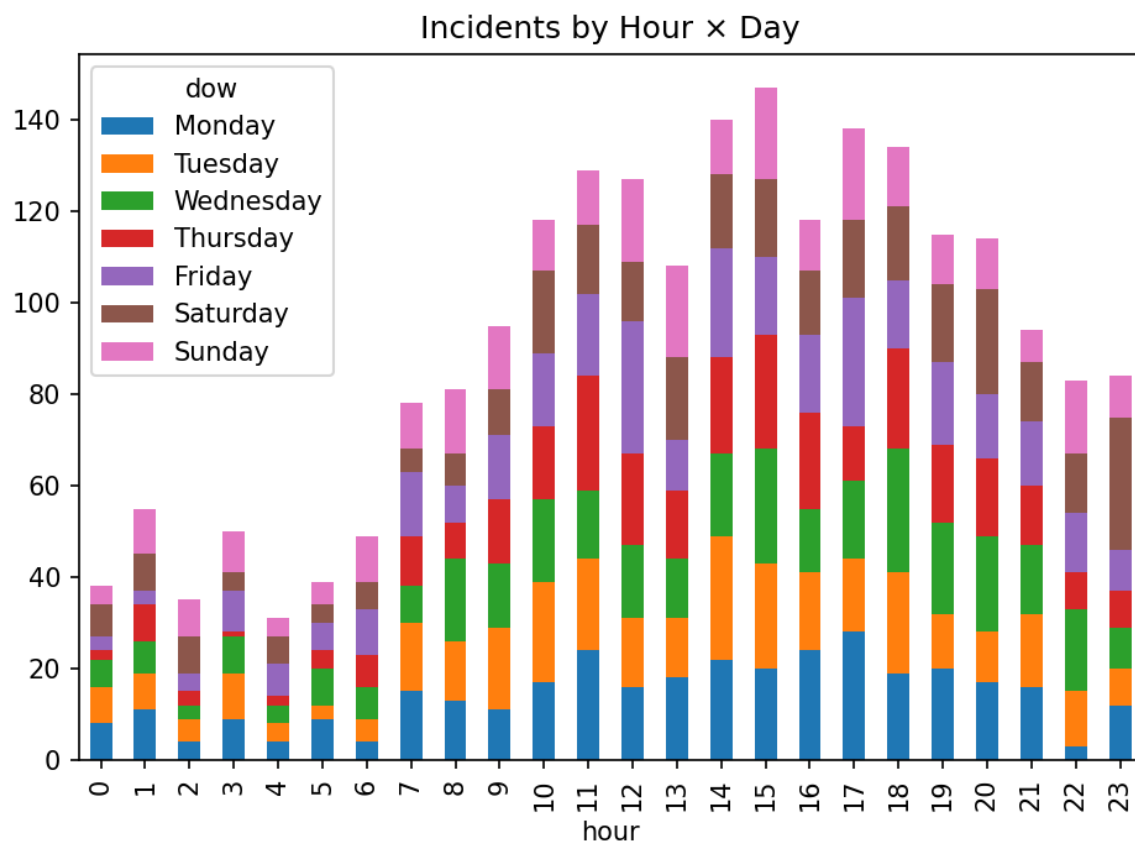


Figure 5: Incidents by hour $\times$ day of week (visualization of the pivot totals).

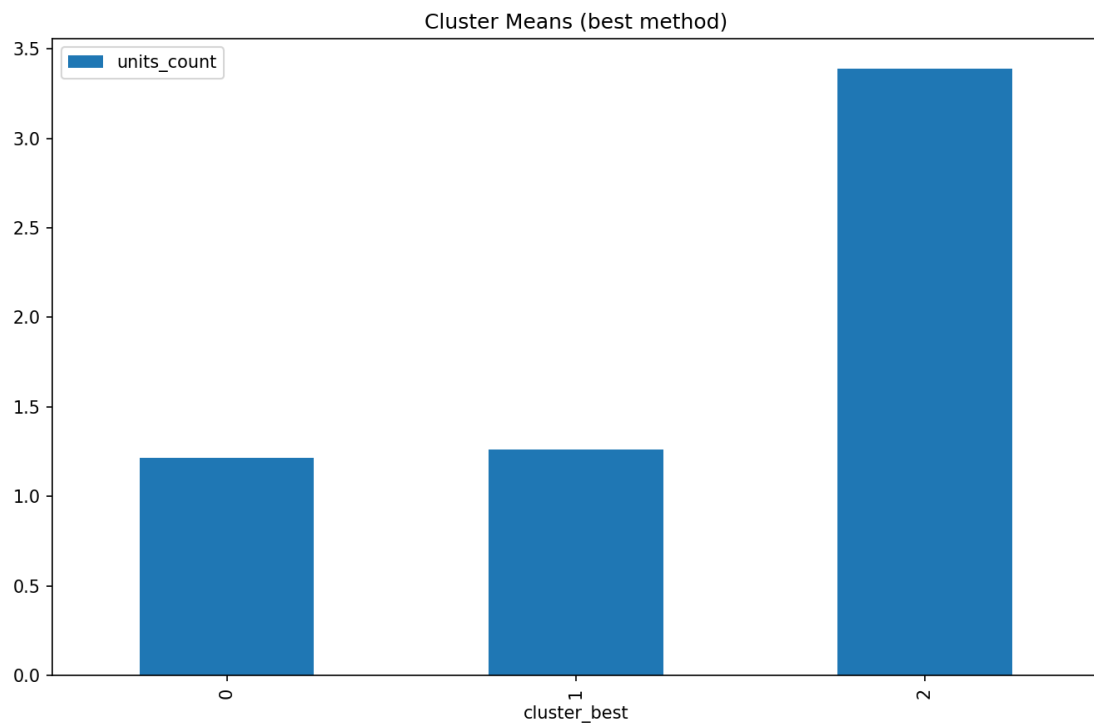


Figure 6: Cluster means for the selected method (see notebook for exact k and silhouette).